Al Isra Private University
Faculty of Engineering
Electrical Engineering Department

Student Name:_____
Student Number:_____

First Semester: 2005/2006
Second Exam

Course Name:	Systems & Signals Processing	Date:	Dec. 21, 2005
Course No:	402-322	ExamTime:	1
Instructor:	Eng. Juma Abu-Ghalune	No. of pages:	1

1. Using Table 4-1 and Table 4-2, find the Fourier Transform of:
   a. $y(t) = 4.\cos(200t).\mathrm{sinc}(4t - 8)$.
   b. $z(t) = 8/|4 + j\pi t|^2$.

2. Using Table 4-1 and Table 4-2, find the Inverse Fourier Transform of:
   a. $X(f) = 8.\exp(j5\pi f)/|4 + j\pi f|^2$.
   b. $W(f) = 8/(4 + j200\pi f)^2$.

3. Use the differentiation theorem to find the Fourier transform of the signal $x(t)$ defined by $x(t) = t$, for $|t| \leq 2$, and $x(t) = 2.\mathrm{sgn}(t)$, for $|t| > 2$.

4. Consider a periodic signal $x(t)$ where $x(t)$ in the fundamental period is given by $x(t) = 2t$, for $0 < t < 2$, $x(t) = 2(2 - t)$, and the period $T = 4$. Using Tables 3-1 and 3-2, find the Fourier series complex coefficients X_n then find the power contained in the first three harmonics.

5. Consider a fixed linear system with amplitude response $3.\Pi[w/600\pi] + 3.\Pi[w/400\pi]$ and phase response $-j0.002\,w$ for $|w| \leq 250\pi$, 0.5π for $w < -250\pi$, and -0.5π for $w > 250\pi$. Find the output and determine the type of distortion for the following inputs:
   a. $w(t) = \cos(250\pi t) + 4\cos(275\pi t)$.
   b. $y(t) = \cos(150\pi t) + \cos(250\pi t)$.

Scanned by CamScanner

1.1 a $y(t) = 4\cos(200t)\,\mathrm{sinc}(4t-8) = 4\cos\left(2\tfrac{100}{\pi}\pi t\right)\mathrm{sinc}(4(t-2))$

$$Y(f) = \frac{1}{2}\left[e^{-j4\pi\left(f-\frac{100}{\pi}\right)}\,\pi\left(\frac{f-\frac{100}{\pi}}{4}\right) + e^{-j4\pi\left(f+\frac{100}{\pi}\right)}\,\pi\left(\frac{f+\frac{100}{\pi}}{4}\right)\right]$$

b. $$z(t) = \frac{8}{|4+j\pi t|^2} = \frac{8}{16+(\pi t)^2} = \frac{4\times 8}{4\times 16+(2\pi t)^2}$$

$$= 2\,\frac{2\times 8}{8^2+(2\pi t)^2}$$

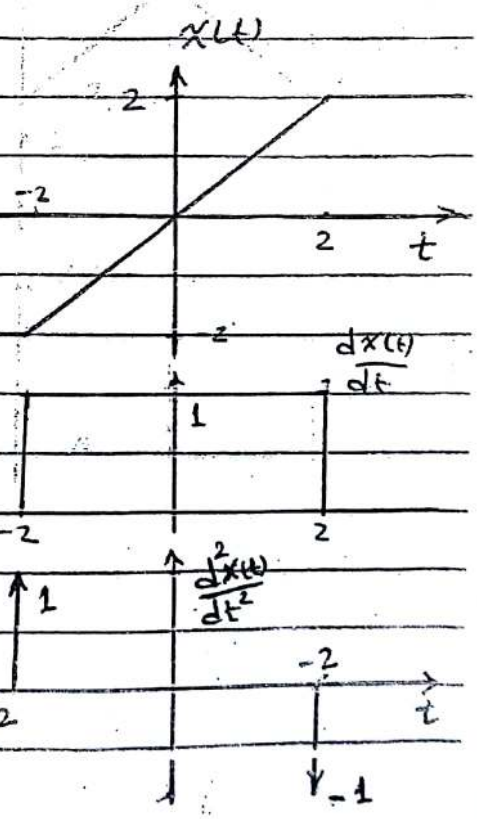
$$Z(f) = 2\,e^{-8|f|}$$

2 a $$X(f) = \frac{8.e^{j5\pi f}}{16+(\pi f)^2} = 2.\frac{2\times 8}{8^2+(2\pi f)^2}\,e^{j2\pi(2.5)f}$$

$$x(t) = 2\,e^{-8|t+2.5|}$$

b. $$W(f) = \frac{8}{(4+j200\pi f)^2} = \frac{8.}{(100)^2[0.04+j2\pi f]^2}$$

$$w(t) = 8\times 10^{-4}\,t\,e^{-0.04t}\,u(t)$$



3 $$\frac{d^2x}{dt^2} = -[\delta(t-2)+\delta(t+2)] = y(t)$$

$$Y(f) = -j2\sin(4\pi f)$$

$$Y(f) = (j2\pi f)^2 X(f)$$

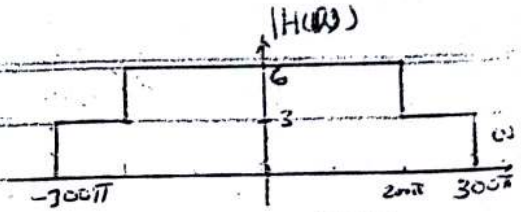
$$X(f) = j\frac{0.5}{(\pi f)^2}\sin(4\pi f)$$

Scanned by CamScanner

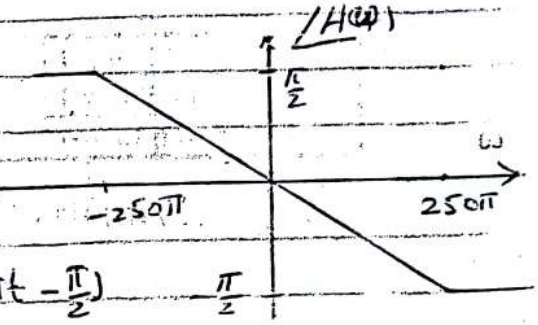
5 a. $u(t) = \cos(250\pi t + 4\cos(2750\pi t)$

$y(t) = 3\cos(250\pi t - \frac{\pi}{2}) + 12\cos($

$275\pi t - \frac{\pi}{2})$.



b There is no distortion



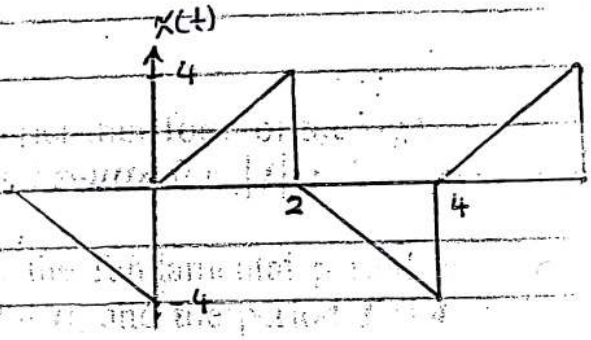
b. $y(t) = \cos(150\pi t) + \cos(250\pi t)$

$y_o(t) = 6\cos(150\pi t - 0.3\pi) + 3\cos(250\pi t - \frac{\pi}{2})$

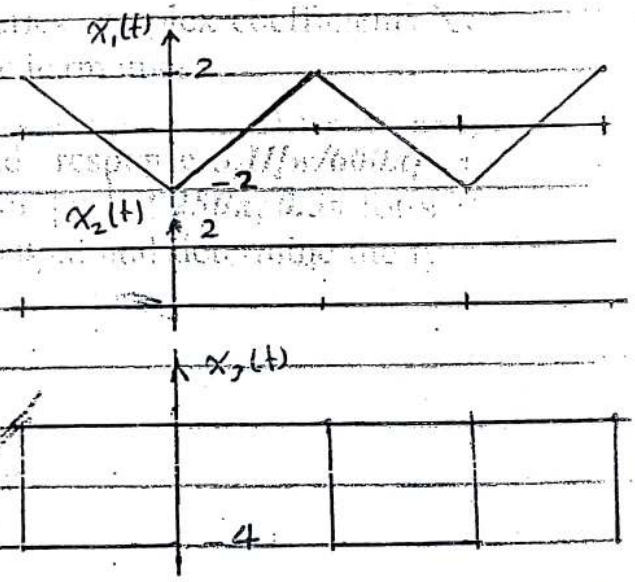
Amplitude distortion

4

$$x(t) = \begin{cases} 2t & 0 < t < 2 \\ 2(2-t) & 2 < t < 4 \end{cases}$$



$x(t) = x_1(t) + x_2(t) + x_3(t)$.



Scanned by CamScanner